

Automated Requirements Engineering in Agile Development: A Practitioners Survey

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Abstract—Requirements Engineering (RE) in the context of agile development is the process of discovering, analysing, validating, and managing software requirements through engaging with the stakeholders. RE activities are central to software development and its success. Advances in automation have been explored in requirements engineering activities and processes in the industry. However, the application of automated support is still limited in practice. In this study, we examine automated RE support in agile development through a qualitative analysis using industry experts. We have found that the application of automated RE support tools is primarily in the experimental stage, largely due to the limited practical implementation in industry settings. The current focus has been on research experiments, with relatively fewer instances of real-world industry application. Although experts are optimistic about its huge potential to reduce development efforts and costs if automation is harnessed. Therefore, we anticipate that these findings will be beneficial for practitioners responsible for developing RE tools for agile development, as well as for the researchers.

Keywords—Requirements Engineering, Automation, Support, Agile Development, Empirical study.

I. INTRODUCTION

The importance of requirements engineering for delivering software products on schedule, within budget, and for the best value has drawn significant attention from both academia and industry [1]. Requirements Engineering, which focuses on the real-world objectives, functionalities, and limitations of software systems, is an important activity in software development [2]. Thus, it has been argued to have a high significance in influencing project results. Over the years, RE has witnessed a transformation from typically a manual process to a more automated process [3] [4] [5]. This is a result of the quest to address the limitations of the traditional RE and in turn reduce development effort, time and cost, which is what agile development seeks to achieve. Automated RE is the use of computer-based tools and techniques to support and improve the process of RE which include eliciting, analysing, specifying, validating, and managing requirements. On the other hand, agile development is characterised by quick, iterative, and incremental development which has become increasingly popular in the last two decades. In agile, development is made lightweight and maintains minimal documentation. As a development methodology, agile introduces its principles and practices that differ from those employed in the more traditional

techniques, including slightly different RE practices referred to as agile RE [1]. These practices have helped to address some of the challenges associated with requirements such as coping with changing requirements as well as poor customer communication.

To enhance communication with customers and facilitate collaboration, several formalization processes have been proposed using automation techniques [6], which are considered integral to the field of automated requirements engineering. Arguably, requirements analysis, developing analysis and design models are time-consuming and complex operations that require automation [7]. The goal of automation in RE is to reduce the time, effort, and cost of the RE process while improving the quality and accuracy of the requirements. This can be achieved through a range of automated techniques, such as Natural Language Processing (NLP), machine learning (ML), and model-based approaches. For example, automated techniques can be used to extract requirements from natural language documents, identify and resolve conflicts among requirements, generate Unified Modeling Language (UML) models from requirements, and verify the completeness and consistency of requirements. These automated techniques can help to improve the efficiency and effectiveness of the RE process, particularly for complex and large-scale software development projects. Thus, these techniques have been utilized to develop tools that automate various activities within requirements engineering, with empirical evaluations conducted on notable examples such as SUGAR [8], CM-Builder [4], DoMoBOT [3], ReqAligner [9], EA-Analyzer [10], REinDetector [11], aTouCan [12], Requirements-Collector [13], and MaraMaic [14], among others.

The industry application of these tools by practitioners has not been reported, specifically in the context of automated requirements engineering in agile development. In order to appreciate these research efforts, particularly the support tools, they need to be applied and used in the industry. This will provide useful feedback to academics and researchers for improvement and potential large-scale deployment. While academic research provides the foundation and initial validation of these tools, their real-world application and feedback from industry practitioners are essential for refining and enhancing their effectiveness. However, efforts have been made to aggregate these studies through a systematic literature review (SLR) to chart a way forward regarding the potential of these tools. Notable examples of such studies include the works by [15] [16] [17]. Despite the availability of these secondary studies, there is a lack of primary studies specifically targeting

the perception of practitioners regarding automated requirements engineering (RE). Therefore, the present study aims to explore automated RE in agile development through a qualitative analysis of practitioners. The main contribution of this work is as follows: to establish the state-of-the-art practices of automated RE among practitioners, and to identify its prospects and challenges.

The remainder of this paper is structured as follows. Section II provides background and related work. Section III describes the approach and methodology of the study. Section IV presents the analysis of the results obtained from the study. The conclusions and future research directions are presented in Section V.

II. BACKGROUND AND RELATED WORK

Research on automated requirements engineering has gained significant attention from researchers in recent times. As a result, several empirical studies have been conducted, proposing various forms of automation. This section provides an overview of earlier work related to RE surveys, highlighting references that discuss the methodology employed or the findings of the studies.

Karlsson *et al.*, [18] conducted an interview-based study with practitioners to understand and describe the challenges of RE in market-driven software development. The study involved semi-structured interviews with fourteen employees from eight software companies, as well as focus group discussions with practitioners. The results of the study identified several challenges associated with RE in market-driven development, primarily of organizational and social nature rather than technical.

Martins and Gorscheck [19] presented the results of a study conducted to examine various aspects of RE in the safety-critical systems (SCS) domain. The study involved interviews with nineteen practitioners who were selected from eleven safety-critical system companies. The results highlighted several findings, including the widespread use of interviews as the primary approach for requirements elicitation, particularly in the SCS domain. In terms of tools used to support requirements specifications and documentation, word processing and spreadsheets were the most commonly utilized, while software engineering tools were used the least. The authors reported that there is a lower adoption of tools in SCS development compared to the traditional software industry. This observation is attributed to the fact that many of the participants interviewed lacked formal education in software engineering or computer science. In fact, only 21% of the practitioners had formal education in these fields, as reported by the authors.

Tehrani *et al.*, [20] conducted an exploratory study aiming to investigate RE activities in model-transformation development. The study employed an interview-based approach with five industry experts, using a semi-structured interview format. The study yielded several conclusions, including the absence of a specific RE process tailored for model transformations. As a result, developers frequently rely on their experience and common sense when applying RE techniques to transformation projects. The study also found that model-transformation projects often function as discrete initiatives

within larger software development endeavours. The relationship between the broader project's needs and the technical nature of model-transformation development was observed to have an impact, particularly on the requirements elicitation process.

Kasauli *et al.*, [21] presented results from multiple case studies to highlight the challenges and practices of requirements engineering in large-scale agile development projects. The results were presented from interviews, focus group discussions, and cross-company workshops which are used for data collection and validation of results. The findings show that; in all cases, the existing co-existence of plan-driven and agile development within the system engineering context limits the possible development pace; strong approaches to RE are required, especially when it comes to documenting a system's behaviour for upcoming feature requests or maintenance. There were 24 challenges identified, which were subsequently divided into six categories and related to certain available solutions. Thus, the study concludes that neither scaled-agile frameworks nor traditional requirements engineering provides sufficient concepts to handle requirements knowledge successfully when working at the scale and pace that the case organisations desire.

Jarzębowski and Weichbroth [1] conducted a systematic literature review and interview study with experienced practitioners on the non-functional requirements in agile development (ASD). The findings reported include a lack of elicitation techniques aimed solely at NFRs, and different documentation technique choices, including the additional techniques introduced to deal specifically with NFRs. In conclusion, it was reported that it was not confirmed that ASD practitioners perceive requirements engineering activities related to NFRs as a problem, as they developed effective practices to address this issue. Furthermore, their findings demonstrate that different approaches to NFRs can be used and produce satisfactory results.

Kabaale and Nabukenya [22] presented findings from an exploratory study of RE process improvement in small and medium-scale enterprises using a systematic approach. The analysis of the findings is based on the data collected from four Ugandan Small and Medium Enterprises (SME) software companies through interviews, questionnaires, and existing requirements documents. Their findings reveal that most of the challenges associated with the RE process come from the size and characteristics of SMEs. These include limited funding, strict deadlines, lack of qualified personnel, management reluctance to support the endeavour and unclear user request that change on a regular basis. Thus, the authors proposed Requirements Engineering Process Improvement (REPI) for SMEs, a systematic strategy that can be learned and understood through a step-by-step process, since most of the case studies have little or no defined RE process.

Svensson *et al.*, [23] conducted a study to identify quality issues in requirements engineering for embedded systems using interview-based study. Data for the study were collected from product managers and project leaders from five software companies. The results show that product usability and performance requirements are considered the two most important types of quality requirements by the practitioners who

were interviewed. Additionally, the authors reported that there were no quality requirements specific elicitation, documentation or analysis, and the companies do not actively look for interdependencies among quality requirements.

Another study by Alsaqaf *et al.*, [24] presented results from a qualitative exploratory case study to examine the challenges of agile quality requirements engineering. The authors conducted eighteen interviews with agile practitioners from six organizations in the Netherlands. The findings identified nine challenges faced by agile practitioners during RE in a large-scale agile project that impede the implementation of the quality requirements. The study added that practitioners either do not understand what quality requirements are or have different ideas; hence, this lack of agreement makes it easy to miss quality requirements. The study concluded that the challenges identified appear not to be caused by Agile but rather those of teams when attempting to implement Agile.

From the foregoing studies, it shows that the qualitative method can be a veritable tool for a study of this nature. Additionally, there is substantial existing work on automated RE which either focus on tools development and evaluation to support the RE process or presents secondary empirical research to aggregate works related to automated RE as mentioned earlier. Thus, there is a need for additional empirical work to understand and establish the perception of practitioners in the industry on the practices, prospects, and challenges of automated RE in Agile development. Since the application of automated RE is relatively low in the industry.

III. RESEARCH APPROACH

To investigate practitioners' viewpoints on automated RE practices, we chose to employ both qualitative and quantitative research approaches. Initially, we conducted a combined quantitative and qualitative study using a structured questionnaire. The questionnaire was distributed to over 200 authors of papers related to automated RE. Specifically, we contacted 217 prospective respondents, with 79.7% representing academics and 20.3% from the industry, accounting for 173 and 44 participants, respectively. The questionnaire consisted of both open-ended and close-ended questions. The first author developed the questions based on the intended information to be collected, and the second author, an experienced researcher and academic, validated them. However, the response rate from the contacted respondents was low. Some respondents began but did not complete the survey and subsequently reached out to us via email, stating that the survey questions were mostly geared towards practitioners, while they themselves were academics. Consequently, due to these responses and the insufficient number of responses for quantitative analysis, we shifted our focus to the qualitative analysis of the available responses. The research process flow is depicted in Figure 1.

There seems to be a lot of work on automated requirements engineering from academia through a lot of research efforts, while some of these publications are authored through a collaborative effort by researchers in the academics and industry, the practical application of these research outputs is yet to be visible. Thus, the few responses we got from the

questionnaire are from the practitioners. The questionnaire comprises close-ended and open-ended questions, which cover a wide range of issues relating to practices in agile development and automated RE.

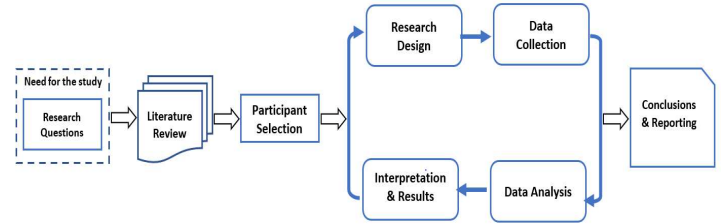


Fig. 1. The flow of the research process.

The findings and analysis reported in this study are based on the data collected through the questionnaires. Altogether, we received 17 completed responses from RE practitioners in the industry, which represents 38.6% of the practitioners contacted. The respondents are experts from different parts of the world. In addition, the respondents were software developers, requirements engineers, project managers as well as software testers. They have participated in different software projects in a wide range of application domains.

Threat to conclusion validity is focused on factors that relate to the treatment and the study's outcome, such as sample size selection, and the care taken in the study implementation and measurement [19]. The variables in our study were examined by using the questionnaire, which included open-ended and close-ended questions in which participants were asked to express their perceptions and points of view. Therefore, threats to validity of conclusions drawn from the questionnaires include: (i) that the respondents are not representative of the automated requirements engineers; (ii) that the questions in the questionnaires were aimed at eliciting a particular response.

Consequently, we tried to avoid the problem (i) by soliciting responses from a wide range of automated RE practitioners. Thus, the candidates for the study were chosen from earlier literature surveys on automated RE. A total of 17 candidates responded to the study, providing valuable insights from a diverse range of organizations. The agile projects covered various domains, including banking, healthcare, government, and services. This diversity in respondents and project domains enhances the richness and applicability of the findings to different industry sectors. Concerning (ii) an expert group reviewed the questionnaire and methodology for ethical approval including checks for bias in questions.

IV. ANALYSIS OF AGILE RE PRACTICES AMONG PRACTITIONERS

In this section, we present our findings concerning the practices of agile requirements engineering from the practitioners' point of view based on the responses provided in the questionnaire.

A. Agile Requirements Engineering

Agile development methods have been widely embraced by software vendors as well as other organizations in the IT sectors during the last two decades [5]. The reasons for the move from

traditional to agile methods suggest that using agile methods results in improved team productivity and confidence, reduced rework, and more effective defect repairs [1]. In the same vein, RE in agile differs from the traditional waterfall approach. While the traditional approach seeks to completely identify the requirements up front by employing sequential practices, requirements in agile are gathered just in time through face-to-face communication sessions with the customer representative who then transmit the elicited requirements to agile teams [24].

In order to understand the extent to which the respondents are employing the use of agile development, we asked questions related to the number of years they have spent working in the software development domain, participation in agile projects, industry sector etc. majority of the respondents have 6 to 20 years experience in the software domain representing 76% of the total respondents. However, they have relatively fewer years of working using automated RE support in software projects. Thus, the highest number of years in automated RE support is 10 years with the majority of the respondent falling within the range of 1 – 5 years which account for 53% of the respondents. Additionally, our study categorized the organizations where respondents participated in agile projects into different size categories, ranging from micro to large, as outlined in Table 1. The findings indicate that a significant portion of the respondents were from medium-sized companies, with employee sizes ranging from 51 to 250 employees.

TABLE I. SIZE OF THE RESPONDENT'S COMPANY

Size	No. of Employee	Frequency	Percentage
Micro	1 – 10	2	7.7
Small	11 – 50	3	15.4
Medium	51 – 250	8	46.2
Large	More than 250	4	30.8

As mentioned earlier, these individuals have played several roles in agile software development projects. The majority of them are project managers, requirements analysts, software designers, and developers/programmers. In some cases, they perform more than one role at a time. While the agile teams they have worked with have at least between 5 to 30 members. From our findings, interview and exploratory prototyping are the commonly used elicitation techniques by practitioners in the industry. Others are brainstorming, scenarios and workshops.

B. Automated RE Practices

As research in automated requirements engineering continues to expand in the academic sphere, its adoption among practitioners in the industry is also gradually increasing. Automated RE has the potential to significantly reduce development efforts, costs, and time. As a result, software analysts are becoming more aware of the benefits that automated support can provide for requirements engineering. To understand the industry practices, we aim to ascertain from the practitioners their awareness and knowledge of some RE tools available in the market, their usage, and the impact of these tools

on development efforts. There is a significant and growing number of RE tools in the market. However, due to the lack of specific capabilities, these tools might not fully support the activities of the RE process, especially considering the complex nature of RE. It is challenging to automate RE activities because they rely heavily on human decision-making [25]. As expressed by multiple respondents, one of whom is quoted as saying, “*There is a need for intelligent computer-aided software engineering (I-CASE) tools*”. Therefore, the advances in artificial intelligence (AI) and machine learning have paved the ways for the development of a system that supports decision making and this has been demonstrated by some of the studies mentioned earlier. Currently, there is the absence of automated tools that will either read the text in natural language or other inputs, analyse it to produce output like UML models or perform other RE related tasks.

Reduction in development efforts is one important factor that both software engineers and customers crave, and this has been the driving force of agile development, especially in reduced development time and fast release to market. Over the years, RE tools have been assisting analysts in the analysis and early design of software products. When asked about the extent to which these tools have reduced development efforts, the majority of the responses indicated a moderate reduction (65%), while 29% reported a high reduction and 6% reported a low reduction. Besides, none of the respondents says that automated support does not reduce development efforts. However, most respondents envision that automated RE tools will have a greater impact on reducing development efforts compared to traditional RE tools. As expressed by one respondent, “*I wish to see I-CASE tools that can facilitate and speed up the process of software development*.” Thus, with the utilization of artificial intelligence, these goals can be realized.

Table 2 shows a selection of the RE tools utilized by practitioners for writing and analysing requirements. Some practitioners rely on word processing applications and spreadsheets to document their requirements. This finding aligns with the observations made by [19], indicating that software requirements are often documented using word processing tools, particularly among individuals lacking a software engineering background or those with limited experience. However, our primary focus is on tools explicitly designed to facilitate RE activities and processes. Based on our findings, modern RE tools appear to be more popular among practitioners in terms of awareness, usage and their impact on reducing development efforts.

C. Automated RE tools evaluation practices

Evaluation is one of the most integral parts of any tool development and automated RE support tools are not an exception. Evaluation is carried out to check for usefulness, completeness, usability, and functionality. Several evaluation methods applied in the industry. Automated RE support tools are designed to perform a wide range of activities such as requirements elicitation (identification) [13], requirements model generation [7] [8] [3], requirements quality analysis [26], and requirements validation [14]. Therefore, from our findings,

case study evaluation is the most widely used (71%), followed by prototyping (proof of concept) and user study representing 59% and 47% respectively. Case study evaluations provide practitioners with firsthand experience and practical insights into the performance, benefits, challenges, and potential of automated RE tools. This empirical approach strengthens the credibility of the tool evaluation process and supports informed decision-making in industry settings. A controlled experiment is the least preferred evaluation method by experts in the industry which account for 35%. This is marginally different from what [15] found in secondary studies which shows that proof of concept is the most widely used in evaluating automated RE support tools.

On the other hand, we found that in the industry there are several evaluation metrics been applied in RE more specifically in the automated tool support. In addition to completeness, correctness, and efficiency measures, other metrics that have to do with process, design, and project metrics are also applied. As a result, combining evaluation metrics and conducting comprehensive evaluations will go a long way toward improving the output of these support tools. Another finding concerning evaluation practices of automated support tools for RE is the absence of an evaluation dataset especially with tools that requires a decision. According, to some of our respondents, most of these tools are evaluated with a particular dataset that is not mostly available to the public. This also makes it difficult for any comparative evaluation analysis to be carried out on the tools.

D. Challenges of automated RE

Requirements engineering is a challenging area considering the complex nature of its activities. There are several research efforts to explore the challenges of RE, while some studies focus on specific domains. For example, safety-critical systems [19], large agile projects [21], and market-driven domains [18]. RE includes several activities undertaken to identify, analyse, validate and management of software and system requirements. Many of these tasks require a significant human effort but could greatly benefit from advances in automation. consequently, we tend to find out the challenges of automated support from the practitioners' point of view.

The automation of RE activities is not matured yet as more research efforts leverage artificial intelligence to deliver improved automation results in RE.

At present, the extracted artifacts or outputs of automated RE activities are not accurate enough to be used in software development either directly or with minor modifications. Therefore, our study identifies several challenges in this regard, which are summarized in Table 3. Accordingly, the top challenge identified by practitioners is the need for domain-specific solutions, with 82% of respondents indicating this as a major concern. This is followed by tool evaluation, as well as requirements traceability and interdependencies, both accounting for 76% of responses. The least challenging aspect of automated RE, according to the survey, is understanding the meaning and semantics of requirements information, which was mentioned by only 29% of respondents.

TABLE II. RE TOOLS

S/N	Tool	Awareness		Usage		Reduced Efforts	
		Freq.*	Percent	Freq.	Percent	Freq.	Percent
1	Modern Requirements (eDev Tech, 4DevOps, Azure DevOps)	12	71	11	65	10	59
2	Jama Software	2	12	2	12	1	6
3	Visure	2	12				
4	IBM Rational DOORS Next	10	59	10	59	9	53
5	IRIS Business Architect	3	18	2	12	2	12
6	Gatherspace	1	6	1	6	1	6
7	Enterprise Architect	9	53	9	53	8	47
8	Agile Manager	6	35	1	6	5	29
9	Tosca Testsuite	1	6				
10	CA Agile Central	2	12	1	6	1	6
11	Agile Designer	7	41	5	29	4	24
12	CodeBeamer	1	6	1	6	1	6
13	Case Spec	1	6	1	6	1	6
14	VersionOne	3	18	3	18	2	12
15	Atlassian Jira	11	65	8	47	8	47
16	RMTrak	1	6	1	6	1	6

- Domain specific solution – In the case of manual RE activities, there are instances where requirements engineers may not have the essential knowledge of the

domain to understand the technical requirements. This is also applicable to automated RE tools. Some of the respondents agree that even though automated RE has a promising potential in addressing the challenges of

elicitation, it is limited because it can only be applied to a known domain and functionalities that can be reused through knowledge bases.

- Processing large volumes of text – most of the RE support tools that were developed are only able to process small text and that is a limitation that needs to be addressed. Requirements are mostly written in natural language in a volume of texts. Therefore, there is a need to improve the tools to be able to process more text.
- Inability to prioritise Requirements – these tools may not be able to prioritise requirements, and this may hamper the project completion time. This challenge is not only peculiar to automated support but also to some practicing software engineers.
- Tool evaluation – in automated RE, tool/output evaluation is a challenge, especially real-world evaluation. The absence of a dataset repository has hindered the effective evaluation of the output of the support tools.
- Absence of identifying non-functional requirements – one important aspect of system requirements is identifying and analysing non-functional requirements. It is observed that the majority of proposed RE automation focuses on functional requirements. Therefore, it will be a great deal of achievement to have support tools to help elicit non-functional requirements.
- Requirements traceability and interdependencies – The inability of the tools to trace the rationale behind certain output decisions is a challenge. Hence, automated tools need to define relationships between requirements and analysis output with considerable coordination efforts. With poor requirements traceability and interdependencies, the software may be difficult to maintain.
- Lack of evaluation dataset – The absence of standardise evaluation datasets is a challenge because every tool uses different datasets which makes it difficult to make comparisons. Automated support tools are supposed to be data-driven, hence, the need for a dataset.
- Stakeholder communication – One of the important aspects of requirements engineering is stakeholder communication. Poor requirements quality is caused by a communication gap throughout the requirements gathering and analysis process between the requirements engineer and stakeholders.
- Coping with changes in requirements – In automated RE it is a challenge to design requirements documents in such a way that they can be adapted to changing surrounding factors at reasonable effort.
- Stakeholder involvement – In automated RE it is a challenge to involve stakeholders throughout the whole development process in regular iterations so that product development will succeed.
- Requirements management – In automated RE continuous management of requirements is a challenge since not all of them are fixed at the beginning and they may change over the course of the project.

- Implicit meaning and semantic information – another limitation of and challenge of automated RE is the understanding of the semantics of words to avoid ambiguity.

TABLE III. CHALLENGES OF AUTOMATED RE SUPPORT

S/N	Challenge	Freq. (n=17)	Percent	RE Phase
1	Domain specific solution	14	82	Elicitation
2	Processing large volume of text	7	41	Analysis
3	Inability to prioritize Requirements	12	71	Elicitation
4	Tool evaluation	13	76	Validation
5	Absence of identifying non-functional requirements	9	53	Elicitation
6	Requirements traceability and interdependencies	13	76	Elicitation
7	Lack of evaluation dataset	12	71	Validation
8	Stakeholder communication	9	53	Elicitation
9	Coping with changes in requirements	11	65	Management
10	Stakeholder involvement	7	41	Elicitation
11	Requirements management	8	47	Management
12	Implicit meaning and semantic information	5	29	Elicitation

V. CONCLUSION AND FUTURE DIRECTION

In this study, we examined the practices of automated requirements engineering in agile development from the perspective of practitioners. The data for this study were collected from 17 industry experts and analysed qualitatively. Requirements engineering is leveraging the advancements in automation, and thus far, the results are promising. However, our findings indicate that practitioners in the industry still rely on existing requirements engineering tools available in the market for specific tasks, despite significant research advancements in tool support for requirements engineering within the industry, particularly in the analysis and specification phase. These tools are designed to assist with tasks such as software requirement modeling (e.g., UML), consistency checking, detecting omissions in requirements, and validating requirements, among others. Based on these findings, it is anticipated that the provision of intelligent computer-aided software engineering tools, incorporating artificial intelligence, machine learning, knowledge-based systems and other techniques, will significantly contribute to reducing development efforts and costs.

However, more research is required to improve the accuracy of the output of the automation by integrating more intelligence into the system. Another research implication for both academia and industry is the establishment of a robust dataset to enhance the evaluation process of support tools. An evaluation dataset is crucial to enable the development of data-driven support tools and fully leverage the advancements in artificial intelligence and machine learning. Lastly, organizations should proactively address the challenges related to automated support and requirements engineering as a whole. This proactive approach will ultimately lead to reduced development costs and efforts.

Regarding the future direction of this work, our intention is to expand the geographical coverage and the sample size of our study and delve into the exploration of automated requirements engineering support practices in specific domains. By broadening the scope of our research, we aim to gain a more comprehensive understanding of the application of automated techniques in different regions and industries. This will allow us to uncover domain-specific insights and address the unique challenges and opportunities that arise in those contexts.

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